Prediction Models and Delay Prediction

This notebook uses the cleaned dataset for:

1. Lead time regression models
2. Delay classification and single-order delay prediction

In [1]:
```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor, RandomForestClassifier
from sklearn.svm import SVR
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
from sklearn.metrics import classification_report, confusion_matrix, accuracy_
```

In [2]:
```
file_path = "/home/pratyush_device/Documents/college/AIML lab/Project/customer_
df = pd.read_csv(file_path)

print("Loaded shape:", df.shape)
df.head()
```

```
Loaded shape: (750, 26)
```

Out[2]:

	acquisition_cost_usd	order_value_usd	satisfaction_score	support_tickets	lead_time_days	days_
0	850	12500	4	1	14	
1	920	8500	5	0	16	
2	880	21000	4	2	15	
3	790	7500	3	1	13	
4	950	15000	5	0	12	

5 rows × 26 columns

In [3]:
```
# Regression target
target_col = "lead_time_days"

if target_col not in df.columns:
    raise ValueError("lead_time_days column not found in cleaned dataset.")

# Convert all non-numeric predictors to numeric for model compatibility
X_reg_raw = df.drop(columns=[target_col])
X_reg = pd.get_dummies(X_reg_raw, drop_first=True)
y_reg = df[target_col]

X_train, X_test, y_train, y_test = train_test_split(
    X_reg, y_reg, test_size=0.3, random_state=40
)

scaler = StandardScaler()
```

```
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

print("Regression features:", X_reg.shape[1])
```

```
Regression features: 25
```

In [4]:

```
# Train regression models
reg_models = {
    "LinearRegression": LinearRegression(),
    "RandomForestRegressor": RandomForestRegressor(random_state=42),
    "SVR": SVR()
}

reg_results = []
for name, model in reg_models.items():
    model.fit(X_train_scaled, y_train)
    preds = model.predict(X_test_scaled)
    reg_results.append({
        "Model": name,
        "MAE": mean_absolute_error(y_test, preds),
        "MSE": mean_squared_error(y_test, preds),
        "R2": r2_score(y_test, preds)
    })

reg_results_df = pd.DataFrame(reg_results).sort_values(by="R2", ascending=Fals
reg_results_df
```

Out[4]:

	Model	MAE	MSE	R2
1	RandomForestRegressor	0.236711	0.246481	0.929357
0	LinearRegression	0.436957	0.336066	0.903681
2	SVR	0.540652	0.609542	0.825301

In [5]:

```
# Create delay label from lead_time_days using 75th percentile threshold
delay_threshold = df[target_col].quantile(0.75)
df["delay"] = (df[target_col] > delay_threshold).astype(int)

print("Delay threshold:", delay_threshold)
print("Class distribution:")
print(df["delay"].value_counts(normalize=True))
```

```
Delay threshold: 16.0
Class distribution:
delay
0    0.877333
1    0.122667
Name: proportion, dtype: float64
```

In [6]:

```
# Delay classification dataset
X_cls_raw = df.drop(columns=[target_col, "delay"])
X_cls = pd.get_dummies(X_cls_raw, drop_first=True)
y_cls = df["delay"]

Xc_train, Xc_test, yc_train, yc_test = train_test_split(
    X_cls, y_cls, test_size=0.2, random_state=42, stratify=y_cls
)

clf = RandomForestClassifier(n_estimators=200, random_state=42)
```

```
clf.fit(Xc_train, yc_train)

yc_pred = clf.predict(Xc_test)
print("Classification features:", X_cls.shape[1])
print("Accuracy:", accuracy_score(yc_test, yc_pred))
print("\nConfusion Matrix:")
print(confusion_matrix(yc_test, yc_pred))
print("\nClassification Report:")
print(classification_report(yc_test, yc_pred))
```

```
Classification features: 25
Accuracy: 1.0

Confusion Matrix:
[[132   0]
 [  0  18]]

Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00       132
           1       1.00      1.00      1.00        18

    accuracy                           1.00       150
   macro avg       1.00      1.00      1.00       150
weighted avg       1.00      1.00      1.00       150
```

In [7]:

```
def get_risk(prob):
    if prob > 0.7:
        return "High"
    elif prob > 0.4:
        return "Medium"
    return "Low"


def recommend_action(risk):
    if risk == "High":
        return "Expedite shipment or change supplier"
    if risk == "Medium":
        return "Monitor order closely"
    return "No action needed"


def predict_order(input_data):
    pred = clf.predict(input_data)[0]
    prob = clf.predict_proba(input_data)[0][1]
    risk = get_risk(prob)
    action = recommend_action(risk)
    return {
        "Delay Prediction": int(pred),
        "Delay Probability": float(prob),
        "Risk Level": risk,
        "Recommended Action": action
    }
```

In [8]:

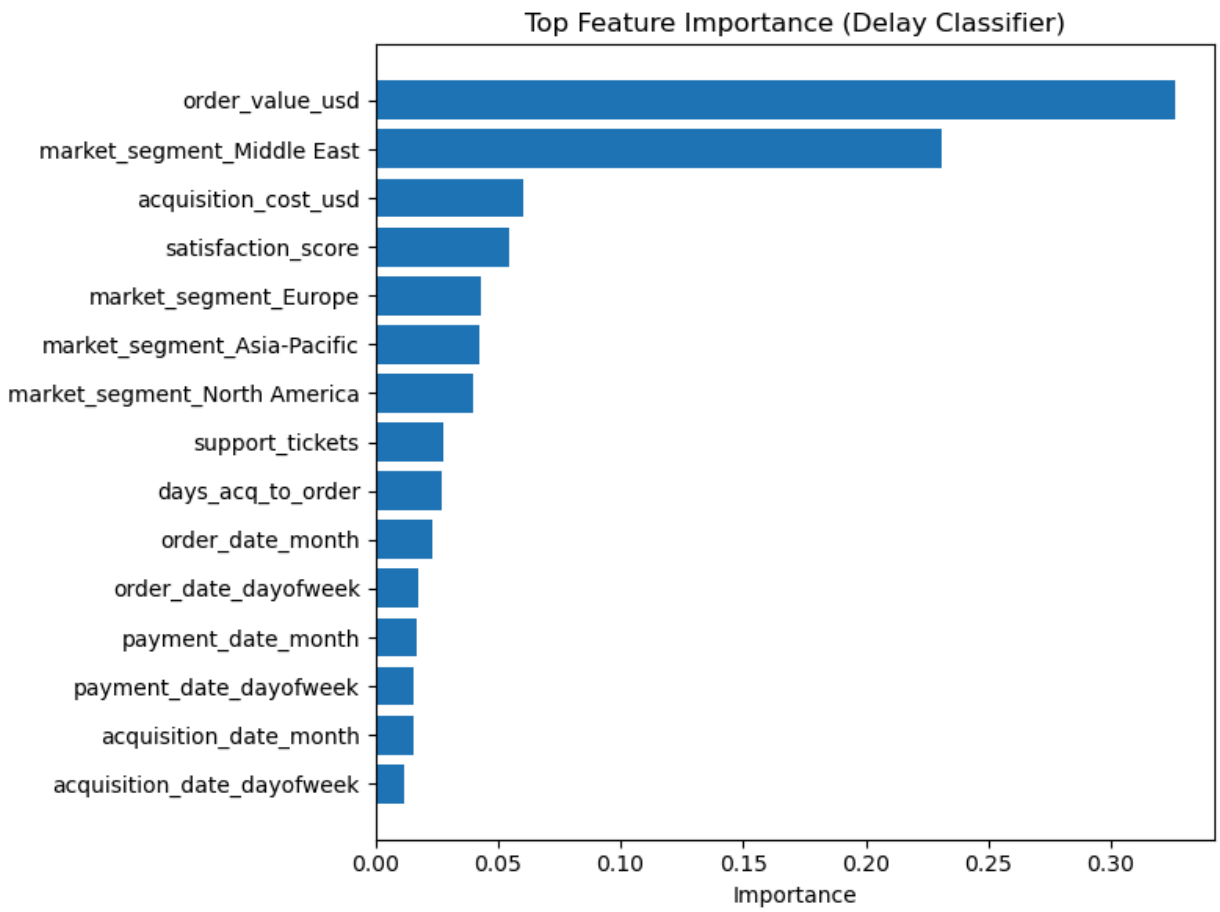
```
# Example prediction for one order from test set
sample = Xc_test.iloc[[0]]
result = predict_order(sample)
result
```

Out[8]:
```
{'Delay Prediction': 0,
 'Delay Probability': 0.06,
 'Risk Level': 'Low',
 'Recommended Action': 'No action needed'}
```

In [9]:
```
# Feature importance for delay classifier
import matplotlib.pyplot as plt

importances = pd.Series(clf.feature_importances_, index=X_cls.columns)
importances = importances.sort_values(ascending=True).tail(15)

plt.figure(figsize=(8, 6))
plt.barh(importances.index, importances.values)
plt.title("Top Feature Importance (Delay Classifier)")
plt.xlabel("Importance")
plt.tight_layout()
plt.show()
```



In [ ]:

In [ ]:

In [ ]: